



Academic Standards for Information Systems Programs

2025

Version 2.0



Introduction

This document aims to contribute to the establishment of minimum requirements for bachelor's degree programs in Information Systems to ensure the academic quality of the programs. It seeks to ensure that graduates are highly qualified, possessing the knowledge, skills, and values required by the labor market, in accordance with national trends and in line with best practices required for the academic and professional pursuit of the specialty.

The Education and Training Evaluation Commission (ETEC), according to its mandate by virtue of the Council of Ministers decision No. 108, dated 14/2/1440 AH, has developed specialized academic standards for Information Systems programs. The remit of the Commission includes building systems for evaluating, approving, and accrediting programs in education and training, including institutional and programmatic, encompassing rules, standards, frameworks, and indicators, and building and developing high-quality national academic programs.

Terms

Academic Standards: Established benchmarks for the quality and level of student achievement in an academic program, outlining the expected knowledge, skills, and values upon completion and ensuring consistency, fairness, and credibility in learning and assessment as a foundation for quality assurance in higher education.

Education and Training Evaluation Commission (ETEC): Independent body aimed at evaluating, assessing, and accrediting qualifications in education and training in both public and private sectors, raising the quality and efficiency of those qualifications and ensuring they contribute to the national economy and development.

Essential Knowledge Units (EKUs): Knowledge Units necessary for future learning in a given discipline.

General Knowledge Units (GKUs): Knowledge Units that should be introduced to students majoring in a discipline.

Key Learning Outcomes (KLOs): The minimum required Learning Outcomes in the discipline that students are expected to obtain.

Knowledge Units (KUs): Mandatory multiple related topics that must be included in an institution's degree program.

Learning Outcomes (LOs): Description of what a learner is expected to know, understand, and be able to achieve, which is represented in his/her behavior and ability at the end of a specific educational program.

National Qualifications Framework (NQF): A comprehensive and uniform structure for building, organizing, and categorizing qualifications into levels based on Learning Outcomes.

Specific Knowledge Units (SKUs): Knowledge Units derived from a General Knowledge Unit.

Specific Learning Outcomes (SLOs): Learning Outcomes for a Specific Knowledge Unit.





Goals

The purpose of this work is to develop the minimum Knowledge Units (KUs) in Information Systems programs, specifying a set of Specific Learning Outcomes (SLOs) that students are expected to acquire after completing their studies. This document lays the foundations for the design of the academic program and the study plans and the alignment of the program with the demands of the labor market. This document also lays out the curriculum design, to facilitate selecting the appropriate teaching strategies and methods and tools for evaluation of the students.

Methodology

This document describes the minimum Essential, General, and Specific Knowledge Units (EKUs, GKUs, and SKUs) in Information Systems programs. The Learning Outcomes (LOs) of each KU set the threshold for what the students are expected to learn and be able to achieve after successfully completing that KU. Educational institutions should take into account the depth and breadth of these KUs so that the LOs integrate communication skills and values into the curriculum. Institutions can also offer additional KUs that are consistent with their objectives. It should also be noted that the KU is not necessarily an independent course; one or more courses can cover a single KU. Likewise, one course could also cover one or more KUs entirely or partially.

The methodology consists of the following phases:

1. Survey and benchmarking

- Benchmarking with international learned societies and professional bodies.
- Benchmarking with top-rated international and local universities.
- Identifying national labor market requirements.
- Incorporating input from specialists and experts from different sectoral groups.

2. Preparation of the contents of the specific standards document

- Identifying program Key Learning Outcomes (KLOs).
- Defining the general characteristics of the curriculum.
- Developing the GKUs and SKUs for each GKU.
- Formulating SLOs for each SKU.
- Determining the minimum topics required for each SKU.
- Describing the methodology for aligning academic content with the National Qualification Framework (NQF).

The KUs are derived from analyzing several high-ranked QS universities and international regulatory bodies and professional associations. The total credit hours outlined in this document account for about 60% of the total credit hours typically required for an Information Systems program.





The ETEC has developed this document in cooperation and coordination with different groups in the field of Information Systems, such as the Ministry of Communications and Information Technology (MCIT), the Saudi Data and Artificial Intelligence Authority (SDAIA), the National Cybersecurity Authority (NCA), the Saudi Information Technology Company (SITE), and the Advanced Technology and Cybersecurity Company (SIRAR by STC), and with the involvement of various stakeholders from government bodies, the private sector, and academia.

Scope and Uses

This document is primarily intended to guide the development of curricula for a bachelor's degree in Information Systems for public and private higher education institutions in Saudi Arabia. It defines the minimum required knowledge, skills, and values that must be incorporated into program design to ensure graduates meet both academic and professional expectations. It is also used to support the creation of standardized tests and inform accreditation processes, ensuring consistency and quality across programs.

Key Learning Outcomes

Key Learning Outcomes¹ describe the essential knowledge, skills, and values that graduates of the Information Systems undergraduate program will be able to demonstrate once they complete the program.

On successful completion of a bachelor's degree in Information Systems, graduates should be able to:

- KLO 1: Specialist Knowledge for Solving Computing Problems: Apply (a) knowledge of computing fundamentals, (b) knowledge of a computing specialization, (c) mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models for defined problems and requirements; and demonstrate the in-depth knowledge and skills for a professional role in a recognized computing specialization.
- KLO 2: Problem Analysis: Identify, formulate, research literature, analyze complex computing problems and conduct investigation through: (a) experiments, (b) analysis and interpretation of data, as well as (c) synthesis of information to provide valid and substantiated conclusions with holistic considerations.

¹ The key learning outcomes are defined according to the Seoul Accord Graduate Attributes (2025).



- KLO 3: Design/ Development of Solutions: Design solutions with own perspectives for complex computing problems, and design systems, components, or processes that meet identified needs with appropriate consideration for public health and safety, information privacy and security, cyber security, life cycle cost analysis, net zero carbon, resource efficiency, culture, society, and environment.
- KLO 4: Modern Tool Usage: Create, select, adapt and apply appropriate techniques, resources, and modern computing tools, including but not limited to simulation, prediction and modelling, with recognition of their limitations, to complex computing activities and problems.
- KLO 5: Individual and Team Work: Participate effectively as an individual and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings.
- KLO 6: Project Management and Finance: Apply knowledge of IT management principles to the work as a member or leader of a team with consideration of market needs and budget, to manage projects under multidisciplinary environments.
- KLO 7: Communication: Communicate effectively on complex computing activities with the computing community and with society at large, such as being able to comprehend and write effective reports, design documentation and make effective presentations, taking into account of cultural, language, and learning differences.
- KLO 8: Computing Professionalism and Society: Analyze and assess the impacts on the sustainable development of professional computing practices and solutions related to the society, economy, information privacy and security, cyber security, health and safety, legal frameworks, corporate governance, entrepreneurship and the environment when solving complex computing problems.
- KLO 9: Ethics: Apply ethical principles, adhere to professional ethics and norms of professional computing practice, accept responsibilities, incorporate diversity and inclusion, and comply with relevant local and international laws.
- KLO 10: Life-long Learning: Recognize the need and have the ability for acquiring independent and life-long learning as a computing professional, adapt to new and emerging computing technologies and exercise critical thinking in the broadest context of technological advancement.

Curriculum General Criteria

Based on the benchmarking study of leading universities and the KUs and skills in the Information Systems programs, the KUs have been grouped into categories. The curriculum should consist of the following:

1. Essential Knowledge Units (EKUs) (minimum of 30 credit hours): Mathematics, Statistics, Basic Sciences, Project Management, and Finance appropriate to the discipline.
2. Computing Knowledge Units:





- 2.1. Computing Knowledge Units (minimum of 40 credit hours).
- 2.2. General Knowledge Units (GKUs): mandatory Information Systems knowledge.
- 2.3. Specific Knowledge Units (SKUs): mandatory Information Systems knowledge derived from GKUs.
3. Professional practice and experiential learning:
 - 3.1. Coverage of ethical aspects of professional practice.
 - 3.2. Minimum of three semester credit hours (or equivalent) for a capstone project.
 - 3.3. Field training experience of a minimum duration of eight weeks.

Each group consists of different subgroups essential in any typical Information Systems program. These subgroups are identified in the next section.

Knowledge Units

Essential Knowledge Units (EKU)

The Essential Knowledge Units should consist of a minimum of 30 credit hours in Mathematics, Statistics, Basic Sciences, and Project Management and Finance. This should include 23 credit hours of Mathematics, Physics, and Chemistry as shown in Table 1. Additional credit hours may be allocated to other Basic Sciences to address specific disciplinary depth requirements. Standardized tests should not include these EKUs.

Table 1: Essential Knowledge Unit for Information Systems programs

#	EKU	Description	Minimum Requirements (credit hours)
1	Mathematics and Statistics	Covers discrete mathematics, calculus, linear algebra, and probability to form the foundation of computing. This ECU equips students with the logical structures and continuous analytical tools necessary for algorithmic analysis, optimization, and statistical inference	11
2	Physics	Provides fundamental knowledge of physical laws, emphasizing mechanics and electromagnetism to understand hardware architecture and physical system constraints. This ECU covers electric circuits and magnetic fields.	6
3	General Chemistry	Provides the scientific basis for understanding material properties, energy conversion, and material constraints of computing hardware, including atomic structure, chemical bonding, and electrochemistry.	4





4	Project Management and Finance	Covers essential professional skills in project management principles, financial planning, budgeting, and risk assessment necessary for leading and justifying complex computing and technology initiatives.	2
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Program Core Knowledge Units

The following table (Table 2) provides an overall view of the curriculum distribution of General and Specific KUs. The table provides general recommendations on the acceptable range of topic distribution for each KU. The weighting of the credits is calculated based on a minimum of 40 GCU credits, as shown in Table 2.

Table 2: General and Specific Knowledge Units of an Information Systems program.

GKU	Weight	SKU	Weight
1. Foundations of Information Systems	30%	1.1. Foundations of information systems	30%
2. Data and Information	10%	2.1. Data and information management	5%
		2.2. Data and business analytics	5%
3. Information System Technology	18%	3.1. Information technology infrastructure	12%
		3.2. Secure computing	6%
4. Application Development	20%	4.1. Systems analysis and design	6%
		4.2. Programming	8%
		4.3. Application development	6%
5. Organizational Ethics and Management	10%	5.1. Information systems ethics and societal impact	4%
		5.2. Information systems management and strategy	6%
6. Information System Integration	12%	6.1. Information systems project management	12%





Appendix A: Alignment of Key Learning Outcomes of an Information Systems Program within the National Qualifications Framework

Table A1 shows how each KLO aligns with specific areas within the National Qualifications Framework (NQF).

Table A1: Alignment of the Key Learning Outcomes for an Information Systems program with the NQF.

Information Systems Key Learning Outcomes	NQF Learning Areas		
	Knowledge and Understanding	Skills	Values, Autonomy, and Responsibility
1.	✓		
2.		✓	
3.		✓	
4.		✓	
5.			✓
6.			✓
7.		✓	
8.			✓
9.			✓
10.	✓		✓





Appendix B: Learning Outcomes and Topics for Knowledge Units

Essential Knowledge Unit (EKU.1): Mathematics and Statistics

Description	Mathematics and Statistics form the rigorous foundation for Computer Science. This EKU encompasses discrete mathematics for analyzing logical structures and algorithmic complexity, alongside continuous analytical tools such as calculus and linear algebra. It integrates probabilistic reasoning and statistical inference necessary for modeling complex systems, processing data, quantifying uncertainty, and deriving robust computing solutions.
Topics	The following topics must be included in this EKU: 1. Discrete Mathematics 2. Calculus 3. Linear Algebra 4. Probability Theory 5. Statistical Inference and Modeling
Specific Learning Outcomes	By completing this EKU, students should be able to: 1. Analyze discrete mathematical structures to solve computing-related problems. 2. Apply logic and proof techniques to validate the correctness of algorithms 3. Analyze the properties of infinite series expansions related to function approximation. 4. Calculate partial derivatives and use optimization methods for gradient-based learning. 5. Perform matrix operations and solve systems of linear equations. 6. Apply counting techniques to compute probabilities for discrete scenarios. 7. Formulate probabilistic models for solving problems involving uncertainty and dependence. 8. Apply hypothesis testing and confidence intervals to draw rigorous inferences about data populations. The table below maps the Specific Learning Outcomes for the EKU to the Key Learning Outcomes.





	SLOs	KLOs									
		1	2	3	4	5	6	7	8	9	10
	1	✓	✓								
	2	✓	✓								
	3	✓									
	4	✓			✓						
	5	✓	✓								
	6	✓									
	7	✓			✓						
	8	✓	✓								





Essential Knowledge Unit (EKU.2): Physics

Description	This EKU provides fundamental knowledge of physical laws, with a focus on Mechanics and Electricity & Magnetism. It covers kinematics, work and energy, electric circuits, and magnetic fields. This unit includes mandatory experimental experience.																																																																																					
Topics	The following topics must be included in this EKU: 1. Mechanics 2. Electricity & Magnetism 3. Optics/Thermodynamics																																																																																					
Specific Learning Outcomes	By completing this EKU, students should be able to: 1. Apply conservation laws and mechanics principles to analyze physical phenomena in computing systems. 2. Analyze the behavior of oscillatory systems and wave propagation phenomena (mechanical and electromagnetic). 3. Analyze performance limitations and failure modes of hardware based on electromagnetic constraints. 4. Utilize principles of thermodynamics to analyze energy transfer, heat management, and cooling needs for high-density computing. 5. Assess thermodynamic cycles to compare operational effectiveness. The table below maps the Specific Learning Outcomes for the EKU to the Key Learning Outcomes. <table> <tr> <th rowspan="2">SLOs</th><th colspan="10">KLOs</th></tr> <tr> <th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th><th>9</th><th>10</th></tr> <tr> <td>1</td><td>✓</td><td>✓</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>2</td><td>✓</td><td>✓</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>3</td><td>✓</td><td>✓</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>4</td><td>✓</td><td></td><td>✓</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>5</td><td></td><td>✓</td><td>✓</td><td></td><td></td><td></td><td>✓</td><td></td><td></td><td></td></tr> </table>										SLOs	KLOs										1	2	3	4	5	6	7	8	9	10	1	✓	✓									2	✓	✓									3	✓	✓									4	✓		✓								5		✓	✓				✓			
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Essential Knowledge Unit (EKU.3): General Chemistry

Description	Chemistry provides the scientific basis for understanding material properties, energy conversion, and device stability. This unit covers the fundamental principles of atomic structure, chemical bonding, and electrochemistry necessary for analyzing material limitations and assessing environmental impact.																																																																																					
Topics	The following topics must be included in this EKU: 1. Atomic Structure and Bonding 2. Stoichiometry and Chemical Reactions 3. Electrochemistry and Energy Systems																																																																																					
Specific Learning Outcomes	By completing this EKU, students should be able to: 1. Describe the relationship between atomic structure, bonding mechanisms, and material periodicity. 2. Relate material structure (molecular/crystal) to macroscopic physical properties. 3. Apply stoichiometry and conservation principles to quantify chemical reactions and mass/energy balances. 4. Analyze energy conversion mechanisms and quantify electrochemical processes. 5. Analyze material and energy considerations to assess environmental impact and inform sustainable design choices. The table below maps the Specific Learning Outcomes for the EKU to the Key Learning Outcomes. <table> <tr> <th rowspan="2">SLOs</th><th colspan="10">KLOs</th></tr> <tr> <th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th><th>9</th><th>10</th></tr> <tr> <td>1</td><td>✓</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>2</td><td>✓</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>3</td><td>✓</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>4</td><td>✓</td><td>✓</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>5</td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td>✓</td><td></td><td></td><td></td></tr> </table>										SLOs	KLOs										1	2	3	4	5	6	7	8	9	10	1	✓										2	✓										3	✓										4	✓	✓									5			✓				✓			
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Essential Knowledge Unit (EKU.4): Project Management and Finance

Description	This unit provides the essential professional knowledge in project management principles, financial planning, budgeting, and economic decision-making, which are explicitly mandated as foundational components of the CS EKU set. This knowledge is crucial for graduates to manage projects and justify solutions under economic and business constraints.																																																																																					
Topics	The following topics must be included in this EKU: 1. Project Management Fundamentals (Project Life Cycle, Planning, Scope, and Scheduling) 2. Risk, Resource, and Budget Management (Risk Identification, Resource Allocation, and Budgeting Principles) 3. Economic Decision Making (Time Value of Money, Feasibility Studies, and Cost/Benefit Analysis)																																																																																					
Specific Learning Outcomes	By completing this EKU, students should be able to: 1. Develop a basic project plan including a defined scope, WBS, and realistic scheduling. 2. Analyze common project risks and allocate resources effectively within a defined budget. 3. Apply budgeting principles to manage project funds and control expenditures throughout the project lifecycle. 4. Evaluate the economic feasibility of technology projects using cost/benefit analysis and time value of money concepts. 5. Apply financial analysis techniques to assess economic viability and return on investment of proposed computing solutions. The table below maps the Specific Learning Outcomes for the EKU to the Key Learning Outcomes. <table> <tr> <th rowspan="2">SLOs</th><th colspan="10">KLOs</th></tr> <tr> <th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th><th>9</th><th>10</th></tr> <tr> <td>1</td><td></td><td></td><td></td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td></tr> <tr> <td>2</td><td></td><td></td><td></td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td></tr> <tr> <td>3</td><td></td><td></td><td></td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td></tr> <tr> <td>4</td><td></td><td></td><td></td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td></tr> <tr> <td>5</td><td></td><td></td><td></td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td></tr> </table>										SLOs	KLOs										1	2	3	4	5	6	7	8	9	10	1						✓					2						✓					3						✓					4						✓					5						✓				
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General Knowledge Unit (GKU) 1: Foundations of Information Systems

Description

This GKU includes topics such as the fundamental concepts of information systems (IS); the collection, processing, storage, distribution, and value of information; and the ability to conduct an organizational business analysis and assess processes and systems.

Specific Knowledge Unit (SKU) 1.1: Foundations of information systems

Description

Students who meet the competencies of Information Systems Foundations can understand the fundamental concepts of IS (including hardware, software, and information acquisition) and the support that IS provides for transactional, decisional, and collaborative business processes. They will also be able to understand the collection, processing, storage, distribution, and value of information and be able to make recommendations regarding IS that support and enable individuals in their daily lives as well as the management, customers, and suppliers of the enterprise. This competency includes the ability to conduct an organizational business analysis and to assess processes and systems.

Topics

The following topics must be included in this SKU:

1. Fundamentals of IS and quality information.
2. IS roles and business decision-making.
3. IS solutions and enterprise systems.
4. IS security, development, and innovation.

Specific Learning Outcomes

By completing this SKU, students should be able to:

1. Classify the components, elements, operations, and impact of IS.
2. Interpret the dimensions, characteristics, and value of quality information.
3. Explain the roles, responsibilities, and characteristics of the IS professional.
4. Recommend techniques for using information and knowledge for business decision-making and strategic value.
5. Analyze common business problems based on different components, elements, types, and levels of information systems.
6. Critique enterprise systems for a given business problem and process.
7. Identify techniques for transmitting and securing information in an organization.
8. Solve basic computational problems using IS development with appropriate methodologies, software tools, and innovative methods for improving processes and organizational change.





The table below maps the Specific Learning Outcomes for the Specific Knowledge Unit to the Key Learning Outcomes.

SLOs	KLOs									
	1	2	3	4	5	6	7	8	9	10
1	✓									
2	✓									
3									✓	
4			✓		✓					
5		✓								
6		✓	✓							
7								✓		
8			✓	✓						





General Knowledge Unit (GKU) 2: Data and Information

Description

This GKU includes topics such as understanding how to gather, organize, curate, and process data to help run an organization or extract actionable information to increase effectiveness.

Specific Knowledge Unit (SKU) 2.1: Data and information management

Description

Students will study tools and techniques for managing data with database systems. At the highest level, two questions are examined: how to use a database and how to build a database. The relational model has been the predominant data management model for over three decades. Most of this module will focus on the classic relational model. In the past several years, driven by an organization's evolving functional and non-functional (quality) needs, alternatives to the classic relational model have emerged. Students will examine illustrative samples of these popular alternatives, known as non-relational or NoSQL models.

Topics

The following topics must be included in this SKU:

1. Relational databases and querying.
2. Programming and securing databases.
3. Concurrency and non-relational models.

Specific Learning Outcomes

By completing this SKU, students should be able to:

1. Formulate queries in relational models using SQL commands to retrieve, manipulate, and manage data effectively.
2. Design relational databases to meet specific requirements, ensuring normalization and integrity constraints.
3. Develop database systems using advanced techniques such as functions and triggers.
4. Implement security measures to protect databases against unauthorized access and threats.
5. Compare the tradeoffs of different concurrency control mechanisms.
6. Develop non-relational data models for specific application needs.

The table below maps the Specific Learning Outcomes for the Specific Knowledge Unit to the Key Learning Outcomes.





	SLOs	KLOs									
		1	2	3	4	5	6	7	8	9	10
	1			✓							
	2			✓							
	3			✓							
	4								✓		
	5		✓								
	6			✓							





Specific Knowledge Unit (SKU) 2.2: Data and business analytics

Description	Students will study principles of computational thinking (CT) to learn data analytics. Analyzing data science problems and expressing it as a data problem is a fundamental part in this module. Different data analytics such as classification, clustering, optimization, recommendation, and visualization are expected to be covered. Students will also learn how to perform exploratory data analysis from inception to the value proposition. This module will cover how to demonstrate the use of big data tools on real world case-studies and navigate around the problems in order to present information in an understandable, efficient, and effective way. The module will also focus on fitting models to produce customized graphs using visualization tools such as excel, Hadoop, Python, and R.
Topics	The following topics must be included in this SKU: 1. Principles of computational thinking and data science. 2. Data analysis and visualization techniques. 3. Big data tools and relational operations. 4. Human perception and aesthetics in data representation.
Specific Learning Outcomes	By completing this SKU, students should be able to: 1. Apply the principles of computational thinking to learn data science. 2. Express a business problem as a data problem. 3. Perform exploratory data analysis from inception to the value proposition. 4. Explain the core principles behind various analytics tasks such as classification, clustering, optimization, and recommendation. 5. Demonstrate the use of big data tools on real-world case studies. 6. Explain the fundamentals of human perception and vision. 7. Present information in an understandable, efficient, effective, and aesthetic manner, for the purposes of explaining ideas and analyzing data. 8. Use data analytics tools such as Hadoop, R, or Python to perform basic data manipulation to produce graphics. The table below maps the Specific Learning Outcomes for the Specific Knowledge Unit to the Key Learning Outcomes.





	SLOs	KLOs									
		1	2	3	4	5	6	7	8	9	10
	1		✓								
	2		✓								
	3		✓								
	4	✓									
	5				✓						
	6	✓									
	7							✓			
	8				✓						





General Knowledge Unit (GKU) 3: Information Systems Technology

Description

This GKU includes topics such as information technology (IT) assets within an organization; its infrastructure and architecture for data, communications, and computing security; and the associated potential adverse consequences within the use domain.

Specific Knowledge Unit (SKU) 3.1: Information technology infrastructure

Description

This area covers all aspects of information technology infrastructure used in the organization. IT infrastructure includes designing and developing suitable architectures or servers, physical and cloud services, capacity planning, and networking. The content covers the installation, configuration, maintenance, and management of all aspects of technology from the server through to the organization's network.

Topics

The following topics must be included in this SKU:

1. IT infrastructure design and architecture.
2. Network principles and configuration.
3. Security and organizational IT infrastructure.

Specific Learning Outcomes

By completing this SKU, students should be able to:

1. Apply key IT infrastructure concepts to design and manage IT infrastructures that meet organizational needs.
2. Design effective IT infrastructure solutions using principles of layered network architectures.
3. Evaluate IT infrastructure solutions utilizing client/server and network hardware (including wireless and wired).
4. Discuss the principles of network software and configuration.
5. Contrast different network protocols and their configurations.
6. Apply network security principles to design and manage secure IT infrastructures.
7. Examine and critique IT infrastructure for organizations.
8. Critique IT server architecture (both physical and cloud-based).
9. Discuss concepts of enterprise architecture.

The table below maps the Specific Learning Outcomes for the Specific Knowledge Unit to the Key Learning Outcomes.





	SLOs	KLOs									
		1	2	3	4	5	6	7	8	9	10
	1			✓							
	2			✓							
	3		✓		✓						
	4	✓									
	5	✓									
	6								✓		
	7			✓							
	8	✓									





Specific Knowledge Unit (SKU) 3.2: Secure computing

Description	This competency area is concerned with practices that ensure secure business operations in the context of adversaries. Assuring secure operations involves creating, operating, defending, analyzing, and testing secure computer systems. Hence, secure computing is an interdisciplinary area, including aspects of computing, law, policy, human factors, ethics, and risk management. The proposed competencies cover these areas but with an IS discipline lens. This includes data security, software security, human security, societal security, and organization security.
Topics	The following topics must be included in this SKU: 1. Data security and cryptography. 2. Human and organizational security. 3. Cybersecurity threats, laws, and standards.
Specific Learning Outcomes	By completing this SKU, students should be able to: 1. Explain the purpose of cryptography in data communications. 2. Apply authentication, authorization, access control, and data integrity concepts to enhance data security. 3. Design secure software to meet important security requirements. 4. Use the concepts of identification, authentication, and access authorization in the context of protecting people and devices. 5. Discuss the importance of social media privacy and security. 6. Apply cyber attack prevention strategies and methods to mitigate their harmful effects. 7. Apply risk management techniques to identify, assess and prioritize risk factors for information assets. 8. Apply security laws, regulations, and standards that govern organizational operations. The table below maps the Specific Learning Outcomes for the Specific Knowledge Unit to the Key Learning Outcomes.





	SLOs	KLOs									
		1	2	3	4	5	6	7	8	9	10
	1	✓									
	2	✓			✓						
	3			✓							
	4	✓							✓		
	5								✓	✓	
	6		✓	✓							
	7		✓								
	8								✓	✓	





General Knowledge Unit (GKU) 4: Application Development

Description

This GKU includes topics such as the development lifecycle of the application or system, and trends in software development.

Specific Knowledge Unit (SKU) 4.1: Systems analysis and design

Description

This SKU examines various systems development methodologies and modeling tools, emphasizing object-oriented systems development methods, software development lifecycles (SDLC), and agile software development while emphasizing analytical techniques to develop the correct definition of business problems and user requirements. Topics should also include design, project management standards, information gathering, effective communication, and interpersonal skill development.

Topics

The following topics must be included in this SKU:

1. Systems development and lifecycle models.
2. Teamwork, roles, and project success.
3. Modeling and design techniques in SDLC.

Specific Learning Outcomes

By completing this SKU, students should be able to:

1. Explain the concepts and processes of systems development, including design, implementation, and evaluation.
2. Demonstrate the SDLC phases and activities.
3. Identify SDLC models (Agile, Waterfall, V-shaped, iterative, spiral, etc.) that are suitable for specific system development cases.
4. Work effectively in a team environment
5. Apply data modeling techniques to design, analyze, and optimize data structures.
6. Describe the role and responsibilities of the participants in the SDLC.
7. Identify common causes of project failure and methods to avoid them.
8. Apply enterprise architecture concepts related to SDLC phases.

The table below maps the Specific Learning Outcomes for the Specific Knowledge Unit to the Key Learning Outcomes.





	SLOs	KLOs									
		1	2	3	4	5	6	7	8	9	10
	1	✓									
	2	✓									
	3		✓	✓							
	4					✓					
	5	✓		✓							
	6						✓				
	7						✓				
	8			✓							





Specific Knowledge Unit (SKU) 4.2: Programming

Description	Programming is the language of computation and logic that sequences and orders instructions to computing hardware in a manner that realizes both correct results and discernable results. Logical structures, algorithms, arithmetic facilities, and the ability to input, store, transform, and output data that can be purposefully used to inform decisions and automated intentional processes are at the heart of learning to program. To program a computer is to meet the computer 'in the middle' such that the growing capabilities of data and computing can be purposefully guided. Programming is meant to shape the mind and reasoning such that human requirements for data and computing outcomes can be expressed and perfected.
Topics	The following topics must be included in this SKU: 1. Foundations of data and programming. 2. Algorithmic problem-solving and program design. 3. Data structures and object-oriented programming.
Specific Learning Outcomes	By completing this SKU, students should be able to: 1. Apply data storage strategies using primitive data types in a computer's volatile memory. 2. Apply data transformations using arithmetic, assignment, and transpositional operators. 3. Write predicate expressions using relational and logical operators. 4. Express algorithmic problem-solving using sequence, selection, and repetition structures. 5. Modularize a program's algorithmic and operating capabilities using functions, methods, subroutines, or similar organizing structures. 6. Apply appropriate linear and non-linear data structures to maintain and manage related data sets in non-volatile memory. 7. Utilize object-oriented concepts in the organization, structuring and implementing behavior and concept management programs. The table below maps the Specific Learning Outcomes for the Specific Knowledge Unit to the Key Learning Outcomes.





	SLOs	KLOs									
		1	2	3	4	5	6	7	8	9	10
	1	✓									
	2	✓									
	3	✓									
	4		✓	✓							
	5			✓							
	6			✓							
	7			✓							





Specific Knowledge Unit (SKU) 4.3: Application development

Description	Application development is the purposeful application of programming fundamentals to craft usable and useful software artifacts and systems to solve actionable business and organizational problems where the power and automation of computing and data processing are warranted. Elements of design, including reconciliation between human social systems and data and information systems, support a software/systems development lifecycle where the industry and craft software realization extend capabilities of software and programming code elements and our understanding of fit and resonance with the human end-users of these systems. In this regard, an information systems perspective on application development, although akin to software engineering, includes human–computer interaction, user experience, and other sociological and psychological components that constitute user and organizational acceptance and satisfaction.
Topics	The following topics must be included in this SKU: 1. Requirements, design, and software architecture. 2. Programming, implementation, and testing. 3. Application deployment and process methodologies.
Specific Learning Outcomes	By completing this SKU, students should be able to: 1. Conduct a systematic requirements analysis to determine the basic facts used to organize the application of programming effort to solve a problem or reach a goal. 2. Formalize and communicate requirements comprehensibly for all stakeholders that will determine the success of the software system. 3. Specify the software system architecture such that the principal components and dependencies of the system are visible and comprehensible for all involved in shaping the materials of design and construction. 4. Identify the lateral components and libraries on which the designed and developed system will depend. 5. Develop the programming code implementation that realizes the system architecture and design. 6. Test all developed programming code components to ensure fidelity, consistency, and fit. 7. Maintain software throughout deployment and utilization to accommodate extant or new intentions and requirements and ensure the intended purpose is well functioning. 8. Adopt an appropriate software systems process methodology so that people, resources, design requirements, and other dynamic considerations allow for correctness and utility.





9. Develop appropriate dialogue among stakeholders to ensure communication and information transparency, maintaining the viability of the software system.

The table below maps the Specific Learning Outcomes for the Specific Knowledge Unit to the Key Learning Outcomes.

SLOs	KLOs									
	1	2	3	4	5	6	7	8	9	10
1		✓								
2							✓			
3		✓	✓							
4		✓	✓							
5		✓	✓							
6		✓	✓							
7						✓				✓
8						✓				
9							✓			





General Knowledge Unit (GKU) 5: Organizational Ethics and Management

Description

This GKU includes topics such as the strategic management of IT in organizations and the practices associated with the ethical use of information systems.

Specific Knowledge Unit (SKU) 5.1: Information systems ethics and societal impact

Description

This SKU covers conceptualizing and implementing environmentally and socially sustainable IT solutions that are aligned with the organization's responsibilities and in compliance with legislative and regulatory requirements and industry standards.

Topics

The following topics must be included in this SKU:

1. Ethical principles and decision-making in IS.
2. Sustainability and responsible IS practices.
3. Data management and stakeholder responsibility.

Specific Learning Outcomes

By completing this SKU, students should be able to:

1. Demonstrate ethical behavior during data collection.
2. Identify the moral issues that surround the storage and usage of data.
3. Compare ethical philosophies and their practical application.
4. Evaluate ethical codes of practice and their implications for society.
5. Identify aspects of sustainability and adaptive systems and data.
6. Categorize ethical stakeholders and their importance to information systems.
7. Apply sustainable processes, actions, and performance to support individuals, organizations, and society.

The table below maps the Specific Learning Outcomes for the Specific Knowledge Unit to the Key Learning Outcomes.





	SLOs	KLOs									
		1	2	3	4	5	6	7	8	9	10
	1									✓	
	2								✓	✓	
	3									✓	
	4								✓	✓	
	5								✓		
	6								✓	✓	
	7								✓		





Specific Knowledge Unit (SKU) 5.2: Information systems management and strategy

Description	IS management competencies cover the capability to develop, maintain, and consistently improve the systems necessary to deliver the information necessary for an organization. The capability focuses on creating value for an organization and on the motivation, performance, and accountability of IS staff. IS strategy emphasizes the competency to create long-term plans for implementing and using organizational information systems to achieve strategic organizational goals and objectives. This area also covers monitoring and controlling organizational IS resources to ensure alignment with and achieving strategies, goals, and objectives.
Topics	The following topics must be included in this SKU: 1. Professional IS management and operational efficiency. 2. Strategic IS planning and alignment with organizational goals. 3. Governance, risk, and compliance in IS management. 4. Workforce development, service management, and governance frameworks.
Specific Learning Outcomes	By completing this SKU, students should be able to: 1. Design and manage an effective IS organization by applying professional managerial skills. 2. Apply operational efficiency and effectiveness in the service delivery of organizational information. 3. Implement strategic plans and manage the information resources in coordination with line management. 4. Develop the oversight mechanisms by which an organization evaluates, directs, and monitors organizational information technology and manages decision-making processes. 5. Evaluate organizational information systems compliance with policies, applicable laws, and regulations. 6. Manage organizational risk and develop risk mitigation plans. 7. Develop IT procurement policies that can be used to negotiate IT contracts. 8. Develop plans for workforce development, training, talent acquisition, and employee retention. 9. Apply leading service management frameworks, such as ITIL and CMMI. 10. Identify commonly used governance frameworks, such as COBIT and TOGAF, to align information systems with organizational requirements. The table below maps the Specific Learning Outcomes for the Specific Knowledge Unit to the Key Learning Outcomes.





	SLOs	KLOs									
		1	2	3	4	5	6	7	8	9	10
	1						✓				
	2						✓				
	3						✓				
	4						✓				
	5								✓		
	6		✓								
	7						✓	✓			
	8					✓					
	9					✓					
	10	✓									





General Knowledge Unit (GKU) 6: Information System Integration

Description

This GKU focuses on integrating and applying knowledge and skills learned across the curriculum through the effective management of information systems projects. Emphasis is placed on project management principles, methodologies, and techniques to ensure successful planning, execution, and delivery of IS projects.

Specific Knowledge Unit (SKU) 6.1: Information systems project management

Description

This SKU provides an understanding of project management concepts and appropriate project management techniques for IS management. Topics include project management principles, project management functions and processes, selecting an appropriate project management methodology, agile software development principles, and scrum. Emphasis is placed on understanding and gaining practical knowledge of essential project management skills: integration management, scope management, time management, cost management, quality management, human resource management, communications management, and risk management. Emphasis is also placed on understanding the scrum process and decision criteria for choosing between planned and agile project management approaches.

Students should also learn the tools, techniques, and processes to manage project performance and move from one phase to another until the project is closed.

Topics

The following topics must be included in this SKU:

1. Fundamentals of project management and lifecycles.
2. Core project management functions and tools.
3. Team, communication, and stakeholder management.
4. Agile and scrum methodologies in IS projects.

Specific Learning Outcomes

By completing this SKU, students should be able to:

1. Identify an appropriate project management methodology based on project characteristics.
2. Use integration and scope management tools, techniques, and processes in IS project management.
3. Develop estimates, time tracking, and cost tracking using appropriate tools, techniques, and processes.
4. Utilize the change control process to maintain and control quality and manage project performance.
5. Implement human resource management tools, techniques, and processes.





6. Implement a communication management plan to manage stakeholders within the phases of a project.
7. Predict and manage project risk through the use of tools, techniques, and processes.
8. Apply agile project management principles and methods in practice.
9. Apply the scrum process, including roles, ceremonies, and artifacts, to deliver projects iteratively.

The table below maps the Specific Learning Outcomes for the Specific Knowledge Unit to the Key Learning Outcomes.

SLOs	KLOs									
	1	2	3	4	5	6	7	8	9	10
1		✓				✓				
2						✓				
3						✓				
4						✓				
5					✓	✓				
6						✓	✓			
7		✓		✓		✓				
8						✓				
9					✓	✓				





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